

Product Information

Injection molding grade with 30% glass fibers for parts requiring enhanced fire resistance (eg relay housings, plug-in connector, switch and lamp parts)

Abbreviated designation according to ISO 1043-1: PBT FR(17)

CLASSIFICATION ACCORDING TO ISO 7792-1:

Moulding Compound ISO 7792-PBT, MFGHLNR, 11-120, GF30

Physical form and storage

Standard packaging includes the 25-kg-bag, the 1000 kg octabin (octagonal container) or the 1000 kg big bag. Other forms of packaging are possible subject to agreement. All containers are tightly sealed and should be opened only immediately prior to processing. Further precautions for preliminary treatment and drying are described in the processing section of the brochure. The bulk density is about 0,7 to 0,8g/cm³.

Ultradur® can be stored for a longer period of time in dry, well vented rooms without causing problems in processing.

Ultradur® should generally have a moisture content of less than 0,04% when being processed.

In order to ensure reliable production, therefore, pre-drying should generally be the rule and the machine should be loaded via a closed conveyor system. Appropriate equipment is commercially available. Pre-drying is also for the addition of batches, e.g. in the case of inhouse pigmentation.

In order to prevent the formation of condensed water, containers stored in unheated rooms must only be opened when they have attained the temperature prevailing in the processing area. This can possibly take a very long time.

Measurements have shown that the interior of a 25-kg bag originally at 5°C had reached the temperature of 20°C in the processing area only after 48 hours.

Product safety

Ultradur® melts are stable at temperatures up to 280°C and do not give rise to hazards due to molecular degradation or the evolution of gases and vapors. Like all thermoplastic polymers, however, Ultradur decomposes on exposure to excessive thermal stresses, e.g. when it is overheated or as a result of cleaning by burning off. At temperatures of > 290 °C can be emitted: carbon monoxide, tetrahydrofuran.

Under special fire conditions traces of other toxic substances are possible. Formation of further decomposition and oxidation products depends upon the fire conditions.

When Ultradur® is properly processed and there is adequate suction at the die no risks to health are to be expected.

Additional safety information can be found in the safety data sheets of the individual products.

Safety data sheets can be requested from the Ultraplaste Infopoint at ultraplaste.infopoint@basf.com.

Note

The data contained in this publication are based on our current knowledge and experience. In view of the many factors that may affect processing and application of our product, these data do not relieve processors from carrying out their own investigations and tests; neither do these data imply any guarantee of certain properties, nor the suitability of the product for a specific purpose. Any descriptions, drawings, photographs, data, proportions, weights etc. given herein may change without prior information and do not constitute the agreed contractual quality of the product. It is the responsibility of the recipient of our products to ensure that any proprietary rights and existing laws and legislation are observed. In order to check the availability of products please contact us or our sales agency.

Product Information

Typical values for uncoloured product at 23 °C ¹⁾	Test method	Unit	Values ²⁾
Properties			
Polymer abbreviation	-	-	(PBT+PET)-GF30 FR(17)
Viscosity number	ISO 307, 1157, 1628	cm ³ /g	108
Processing			
MVR, ASTM	ASTM D 1238	cm ³ /10min	4.98
Temperature, ASTM	ASTM D 1238	°C	250
Load, ASTM	ASTM D 1238	kg	2.16
Shrinkage (Flow direction)	ASTM D 955	%	0.4
Shrinkage (Across flow direction)	ASTM D 955	%	1
Flammability			
Flammability at 0.75 mm thickness	UL-94, IEC 60695	class	V-0
Yellow Card available	UL-94, IEC 60695	-	yes
Mechanical properties			
Flexural strength (ASTM)	ASTM D 790	MPa	224
Flexural modulus (ASTM)	ASTM D 790	MPa	10800
Izod notched impact strength ASTM D 256 (23°C)	ASTM D 256	J/m	99
Tensile strength 5 mm/min (ASTM)	ASTM D 638	MPa	143
Tensile elongation 5 mm/min (ASTM)	ASTM D 638	%	2.6
Modulus of elasticity (ASTM)	ASTM D 638	MPa	11400
Thermal properties			
HDT B (1.82 MPa), ASTM	ASTM D 648	°C	203
HDT B (0.45 MPa), ASTM	ASTM D 648	°C	220
HDT A (1.80 MPa)	ISO 75-1/-2	°C	195
Electrical properties			
Relative permittivity (1 MHz)	IEC 62631-2-1	-	3.7
Dissipation factor (1 MHz)	IEC 62631-2-1	E-4	158
Volume resistivity	IEC 62631-3-1	Ohm*m	1E14
Surface resistivity	IEC 62631-3-2	Ohm	>1E15
Comparative tracking index, CTI, test liquid A	IEC 60112	-	-

Footnotes

1) If product name or properties don't state otherwise.

2) The asterisk symbol "*" signifies inapplicable properties.

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